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An Explainable Artificial Intelligence Framework for Cross-Population Prediction and Fairness Analysis of Chronic Kidney Disease

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Abstract: Chronic kidney disease (CKD) prediction models may perform well internally but fail to generalize across populations. This study proposes an explainable machine learning framework for cross-population and age-group evaluation. The UCI CKD dataset was used for model development and BD-KDD for external validation. Seven clinically comparable variables were harmonized. Logistic regression, random forest and XGBoost were trained using an 80:20 stratified split. Random forest achieved an internal ROC-AUC of 0.9831, which decreased to 0.5290 externally, producing a gap of 0.4541. External permutation analysis identified creatinine and urea as the most influential variables, with importance values of 0.0233 and 0.0112. Age-group analysis showed recall, accuracy and F1-score gaps of 0.1461, 0.0432 and 0.0713, respectively. These findings demonstrate that internal performance may overestimate cross-population reliability and highlight the importance of external validation, feature analysis and subgroup evaluation.

Keywords: chronic kidney disease; machine learning; external validation; cross-population generalizability; explainable artificial intelligence; fairness; BD-KDD

1. Introduction

Chronic kidney disease is a major clinical problem for which early identification and appropriate management are important (KDIGO, 2024). Machine learning has therefore been increasingly applied to structured clinical and electronic health-record data for CKD prediction (Rizwan *et al.*, 2026).

A major methodological problem is that a model can perform well on a random test split but fail when applied to a different population. Prediction-model reporting guidelines emphasize the importance of transparent model development and validation, while independent valida-

tion provides stronger evidence of generalizability than internal testing alone (Collins et al., 2015). This issue is particularly important when datasets differ in demographic composition, measurement distributions or clinical setting.

Existing CKD studies have mainly emphasized predictive performance and model-specific explainability. For example, recent CKD work has investigated multiple ML models together with SHAP, LIME and statistical validation (Rizwan et al., 2026). The present study instead focuses on a different question: *does a model that performs strongly on one CKD population remain reliable on an independent population?*

The study therefore introduces a cross-population evaluation pipeline combining internal evaluation, independent external validation, quantitative internal-external performance gaps, external permutation feature importance and age-group fairness analysis. Permutation importance is particularly suitable because it measures the reduction in predictive performance after disrupting one feature while keeping the fitted model unchanged (Fisher et al., 2019).

The main contributions are:

1. **Cross-population generalizability analysis:** The trained models are evaluated on a completely independent BD-KDD population rather than treating another random split as external evidence.
2. **Internal-external performance gap analysis:** The study explicitly quantifies the degradation between internal and external performance rather than reporting the two evaluations independently.
3. **External feature importance:** Permutation feature importance is calculated on BD-KDD to identify which clinical variables remain influential after population transfer.
4. **Age-group fairness analysis:** External performance is separately evaluated for < 40 , $40-60$ and > 60 years, and the maximum subgroup difference is reported for accuracy, recall and F1-score.
5. **Failure-oriented interpretation:** Instead of interpreting high internal accuracy as evidence of clinical readiness, the framework explicitly investigates whether the model remains discriminative after population shift.

2. Materials and Methods

This study developed an explainable machine learning framework for cross-population CKD evaluation.

2.1. Overall framework

The proposed framework consists of six computational stages:

1. dataset harmonization;
2. preprocessing and stratified internal splitting;
3. model training;
4. independent external validation;
5. external permutation feature importance;

6. age-group fairness analysis.

The complete computational flow is:

$$D_s \rightarrow H(D_s, D_e) \rightarrow D_{train}, D_{test} \rightarrow \{LR, RF, XGB\} \rightarrow P_e \rightarrow \{\Delta_{performance}, PI, FA\}$$

where D_s denotes the source CKD dataset, D_e denotes the external BD-KDD dataset, H denotes feature harmonization, P_e denotes external prediction, PI denotes permutation importance and FA denotes fairness analysis.

2.2. Datasets

The UCI Chronic Kidney Disease dataset contains 400 records and 24 clinical features [?]. It was used only for model development and internal testing.

The BD-KDD dataset contains 988 anonymized patient records and 26 variables, including 481 healthy and 507 kidney-disease cases [?]. It was treated as an independent external population and was not used for training.

Table 1: Datasets used in the cross-population experiment.

Dataset	Records	Original features	Healthy	Disease
UCI CKD	400	24	–	–
BD-KDD	988	26	481	507

The external dataset was deliberately preserved as an independent validation population. This separation is central to the proposed cross-population analysis.

2.3. Feature harmonization

Only variables available in comparable form in both datasets were retained.

Table 2: Harmonized variables used for cross-population prediction.

Common feature	UCI CKD	BD-KDD
Age	age	age
Haemoglobin	haemoglobin	hemo
Hematocrit	hematocrit	pcv
RBC count	rbc_count	rbcc
WBC count	wbc_count	wbcc
Creatinine	creatinine	sc
Urea	urea	bu

The harmonized feature vector is

$$X = [age, haemoglobin, hematocrit, rbc, wbc, creatinine, urea].$$

The target variable was encoded as

$$y = \begin{cases} 0, & \text{Healthy,} \\ 1, & \text{Kidney disease.} \end{cases}$$

2.4. Preprocessing

Numerical variables were converted to numeric representation and missing values were handled using median imputation inside the model pipeline.

The source dataset was divided using an 80:20 stratified split:

$$D_s = D_{train} \cup D_{test}, \quad D_{train} \cap D_{test} = \emptyset.$$

The split preserved the class distribution and used random state 42.

No BD-KDD observations were used during model fitting.

The WBC distributions were also checked because the mean WBC value differed between the populations:

$$\bar{x}_{UCI} = 7.7954$$

and

$$\bar{x}_{BD-KDD} = 9.5799.$$

This difference was retained as part of the population-shift investigation rather than artificially forcing the two populations to have identical distributions.

2.5. Machine learning models

Three classifiers were evaluated.

Logistic Regression was used as a linear baseline:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}.$$

Random Forest combines multiple decision trees:

$$\hat{P}(y = 1|x) = \frac{1}{T} \sum_{t=1}^T I(h_t(x) = 1),$$

where T is the number of trees and h_t is the prediction of tree t [?].

The implemented Random Forest contained 500 trees with balanced class weights.

XGBoost was used as a gradient-boosting model. Its additive form is

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i),$$

where each f_k is a regression tree [?].

The XGBoost configuration used 400 estimators, maximum depth 4, learning rate 0.03, subsampling 0.9 and column subsampling 0.9.

2.6. Evaluation metrics

For binary classification:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}. \quad (4)$$

ROC-AUC was used to measure discrimination across classification thresholds.

The internal evaluation was performed on D_{test} , whereas the external evaluation was performed on the independent BD-KDD dataset.

2.7. Internal-external performance gap

For a metric M , the cross-population degradation was calculated as

$$Gap_M = M_{internal} - M_{external}. \quad (5)$$

For the Random Forest ROC-AUC:

$$Gap_{AUC} = 0.9831 - 0.5290 = 0.4541.$$

This quantity is one of the main outcomes of the study because it directly measures the difference between apparent internal performance and independent population performance.

2.8. External permutation feature importance

The selected Random Forest was evaluated on BD-KDD using permutation feature importance. For feature j , its importance was calculated as

$$PI_j = S(X, y) - \frac{1}{R} \sum_{r=1}^R S(X^{\pi_{j,r}}, y), \quad (6)$$

where S is ROC-AUC, $X^{\pi_{j,r}}$ represents the dataset after permutation of feature j in repetition r , and R is the number of repetitions.

A larger positive PI_j indicates a larger decrease in external ROC-AUC when that feature is disrupted. Permutation importance is model-agnostic and can be estimated repeatedly to quantify variability [?].

Importantly, the analysis was performed on the **external BD-KDD population**, not only on the training data.

2.9. Age-group fairness analysis

Age was divided into three groups:

$$G_1 = < 40, \quad G_2 = 40-60, \quad G_3 = > 60.$$

For metric M , the age-group gap was calculated as

$$Gap_M^{age} = \max_{g \in G} M_g - \min_{g \in G} M_g. \quad (7)$$

The analysis therefore evaluates whether aggregate external performance conceals subgroup differences. Subgroup performance analysis is an established component of healthcare fairness evaluation [? ?].

This study reports fairness descriptively rather than claiming that a single metric proves overall algorithmic fairness.

2.10. Proposed algorithm

Algorithm 1. Cross-population CKD prediction and fairness analysis

1. Input source dataset D_s , external dataset D_e .
2. Select the seven common clinical variables: age, haemoglobin, hematocrit, RBC, WBC, creatinine and urea.
3. Harmonize variable names and target labels between D_s and D_e .
4. Convert selected variables to numerical representation.
5. Apply median imputation within the training pipeline.
6. Split D_s into D_{train} and D_{test} using an 80:20 stratified split.
7. Train Logistic Regression, Random Forest and XGBoost using D_{train} .
8. Evaluate all models on D_{test} using accuracy, precision, recall, F1-score and ROC-AUC.
9. Select the best model based on internal performance.
10. Apply the selected trained model directly to D_e without retraining.
11. Calculate external accuracy, precision, recall, F1-score and ROC-AUC.
12. Calculate the internal-external performance gap for each metric.
13. For every feature j , randomly permute j in D_e and calculate the decrease in external ROC-AUC.
14. Rank features according to mean permutation importance.
15. Divide D_e into < 40 , $40-60$ and > 60 age groups.
16. Calculate accuracy, precision, recall and F1-score for every age group.
17. Calculate the maximum-minus-minimum subgroup gap.
18. Output model results, external validation results, feature importance, age-group results and performance gaps.

3. Results

3.1. Internal model performance

The three models were first evaluated on the internal UCI CKD test set.

Table 3: Internal performance on the UCI CKD test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9000	0.9762	0.8542	0.9111	0.9902
Random Forest	0.9500	0.9231	1.0000	0.9600	0.9831
XGBoost	0.9375	0.9388	0.9583	0.9485	0.9811

Logistic Regression obtained the highest internal ROC-AUC of 0.9902. However, Random Forest obtained the highest internal accuracy, recall and F1-score and was selected as the primary model for the external analysis.

3.2. External validation

The trained models were evaluated without retraining on the independent BD-KDD dataset.

Table 4: External performance on the independent BD-KDD population.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.5010	0.5123	0.5740	0.5414	0.4961
Random Forest	0.5395	0.5389	0.7101	0.6128	0.5290
XGBoost	0.5263	0.5301	0.6765	0.5945	0.5283

Random Forest achieved the strongest external ROC-AUC among the three models, with a value of 0.5290. Its external accuracy was 0.5395, precision was 0.5389, recall was 0.7101 and F1-score was 0.6128.

The external ROC-AUC values of all three models were close to 0.5, indicating that their ranking ability deteriorated substantially after population transfer.

3.3. Cross-population performance gap

Table 5: Internal-external performance comparison.

Model	Internal ROC-AUC	External ROC-AUC	ROC-AUC Gap
Logistic Regression	0.9902	0.4961	0.4941
Random Forest	0.9831	0.5290	0.4541
XGBoost	0.9811	0.5283	0.4528

The Random Forest model showed an internal ROC-AUC of 0.9831 but an external ROC-AUC of only 0.5290:

$$\Delta AUC_{RF} = 0.9831 - 0.5290 = 0.4541.$$

Thus, approximately 0.4541 ROC-AUC points were lost when the model was transferred to the independent BD-KDD population.

For Random Forest, the other performance gaps were

$$\Delta Accuracy = 0.9500 - 0.5395 = 0.4105,$$

$$\Delta F1 = 0.9600 - 0.6128 = 0.3472.$$

These results provide direct evidence that internal performance substantially overestimated external performance.

3.4. External confusion matrix

For the Random Forest model, the external confusion matrix was

$$CM = \begin{bmatrix} 173 & 308 \\ 147 & 360 \end{bmatrix}.$$

Here, 173 healthy records were correctly classified as healthy, while 308 healthy records were incorrectly classified as kidney disease. Among the 507 kidney-disease records, 360 were correctly identified and 147 were misclassified as healthy.

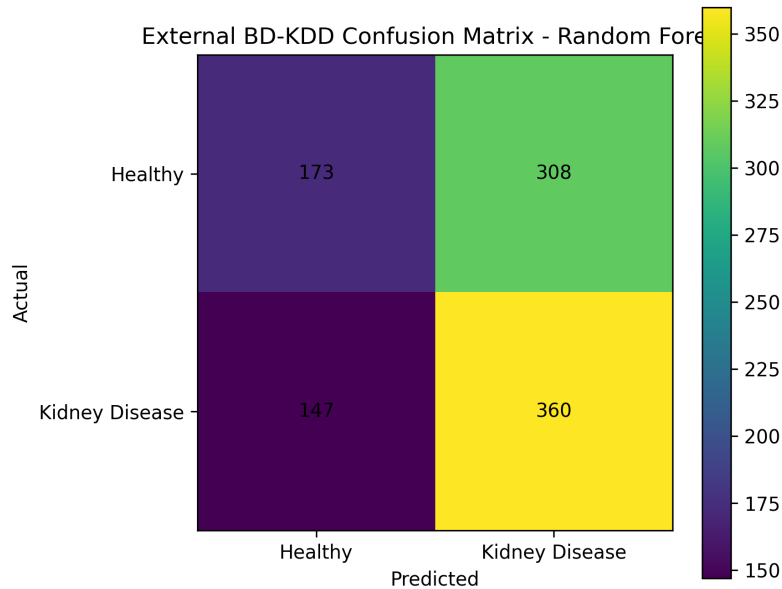


Figure 1: External BD-KDD confusion matrix of the Random Forest model.

3.5. External permutation feature importance

External permutation importance was calculated using Random Forest and the independent BD-KDD population.

Table 6: External permutation feature importance of Random Forest.

Rank	Feature	Mean decrease in ROC-AUC
1	Creatinine	0.0233
2	Urea	0.0112
3	Haemoglobin	0.0016
4	Age	0.0002
5	RBC count	-0.0024
6	WBC count	-0.0032
7	Hematocrit	-0.0054

Creatinine produced the largest external performance decrease:

$$PI_{creatinine} = 0.0233.$$

Urea was the second most influential feature:

$$PI_{urea} = 0.0112.$$

The negative permutation values for RBC count, WBC count and hematocrit indicate that their random permutation did not decrease ROC-AUC in this external experiment. Such values should not be interpreted as evidence that these variables are biologically unimportant; permutation importance measures dependence of the fitted predictive model on the feature under the evaluation distribution [?].

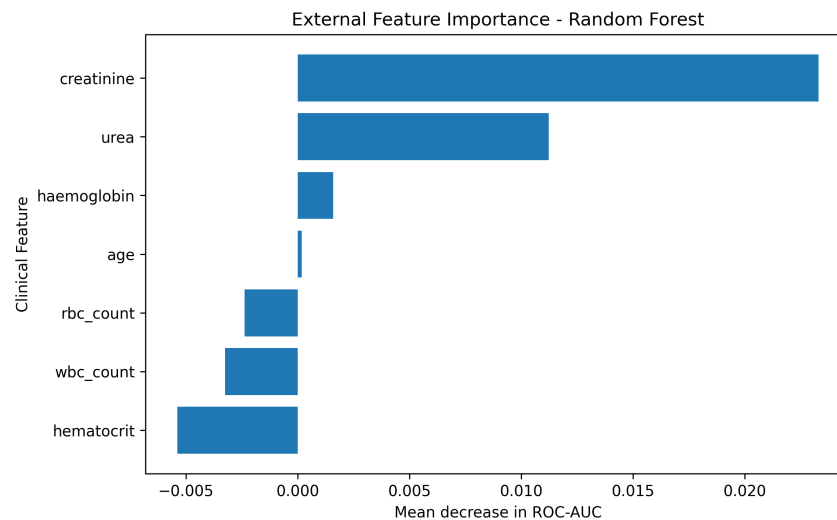


Figure 2: External permutation feature importance of the Random Forest model.

3.6. Internal age-group analysis

The internal test set contained 80 observations.

Table 7: Internal age-group performance.

Age group	N	Accuracy	Precision	Recall	F1
< 40	22	1.0000	1.0000	1.0000	1.0000
40–60	28	0.8929	0.8500	1.0000	0.9189
> 60	30	0.9667	0.9444	1.0000	0.9714

The internal age-group results appear highly favorable, with recall equal to 1.0000 for all three groups.

However, this result must be interpreted together with the much larger and independent external age-group analysis.

3.7. External age-group fairness analysis

Table 8: External age-group performance on BD-KDD.

Age group	N	Accuracy	Precision	Recall	F1
< 40	314	0.5637	0.5639	0.7711	0.6514
40–60	357	0.5350	0.5142	0.7341	0.6048
> 60	317	0.5205	0.5412	0.6250	0.5801

The largest external recall was observed in the < 40 group:

$$Recall_{<40} = 0.7711.$$

The lowest external recall occurred in the > 60 group:

$$Recall_{>60} = 0.6250.$$

Therefore,

$$Gap_{Recall}^{age} = 0.7711 - 0.6250 = 0.1461.$$

Similarly,

$$Gap_{Accuracy}^{age} = 0.5637 - 0.5205 = 0.0432,$$

and

$$Gap_{F1}^{age} = 0.6514 - 0.5801 = 0.0713.$$

These values indicate measurable differences in model performance across age groups. Such subgroup analysis is important because overall performance can hide differences between patient groups [? ?].

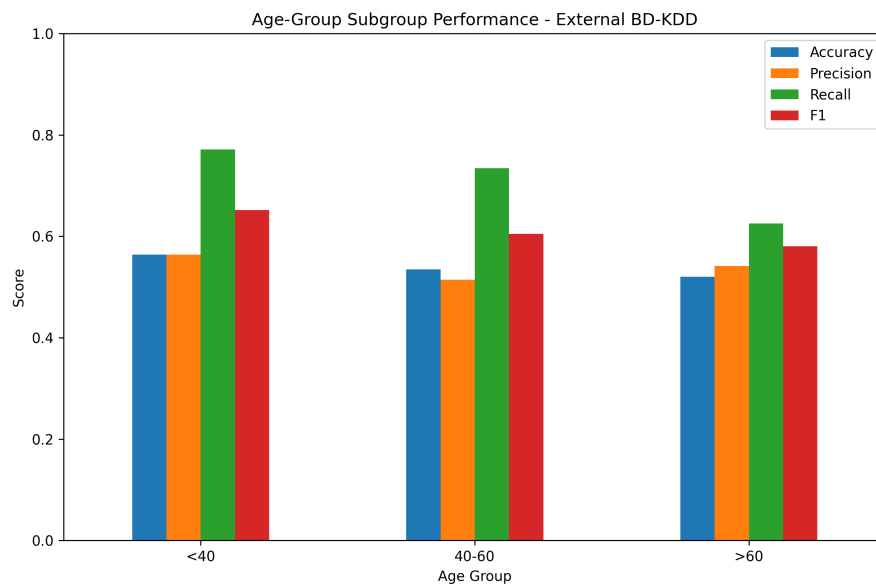


Figure 3: External age-group performance of the Random Forest model.

4. Discussion

The main finding was the substantial reduction in Random Forest performance from an internal ROC-AUC of 0.9831 to 0.5290 on the independent BD-KDD dataset, demonstrating limited cross-population generalizability (Collins et al., 2015). External permutation analysis identified creatinine (0.0233) and urea (0.0112) as the most influential variables. Age-group analysis further showed an external recall gap of 0.1461, indicating non-uniform subgroup performance. These findings highlight the importance of external validation, model explanation and subgroup analysis beyond internal performance alone (Huang et al., 2024).

5. Limitations

The analysis was restricted to seven harmonized variables and three age groups. The relatively small source dataset may also affect internal performance estimates. In addition, the fairness analysis was descriptive and cannot establish the causes of the observed subgroup differences. Further validation using larger and independent populations is required.

6. Conclusions

This study demonstrated that strong internal CKD prediction may not generalize to an independent population. Random Forest achieved an internal ROC-AUC of 0.9831 but only 0.5290 on BD-KDD, indicating a cross-population gap of 0.4541. External analysis identified creatinine and urea as influential features, while age-group evaluation revealed a maximum recall gap of 0.1461.

Overall, the findings highlight the importance of external validation, explainability and subgroup evaluation before clinical deployment.

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Data availability: The source code and supplementary materials are available on (GitHub: Jubaida-78).

Ethical considerations

This study used publicly available datasets. No personally identifiable information was accessed by the authors. The BD-KDD dataset was reported by its creators as anonymized retrospective clinical data obtained with institutional authorization [?]. No additional participant recruitment or intervention was performed in the present computational study.

Conflict of interest

None to declare.

Authors' contribution

Jubaida Begum – conceptualization, data preprocessing, machine learning implementation, cross-population analysis, visualization and manuscript preparation.

Habibur Rahman – literature review, data analysis, interpretation of results and manuscript review.

Anamika Roy – methodology, interpretation of results, manuscript review and editing.

Muhammad Sakib Ibtidha Chowdhury – supervision, research design, methodological guidance, interpretation of findings and final manuscript review.

All authors have read and approved the final manuscript.

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